

Excel Power Pivot

Objectives

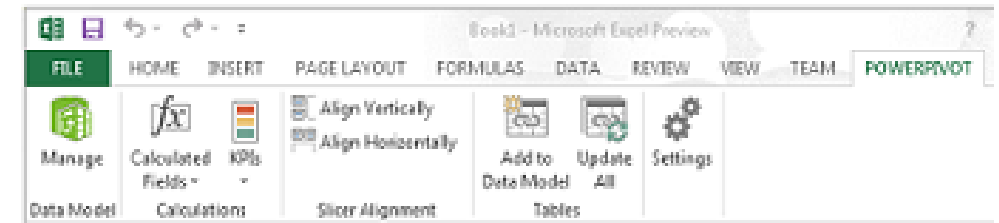
- Summarize data using Microsoft Excel Pivot
- Analyze data using Microsoft Excel Pivot
- Explore data using Microsoft Excel Pivot
- Present data using Microsoft Excel Pivot

Power Pivot

- **Power Pivot** is a data modeling technology that enables you to create data models, establish relationships, and perform calculations. With Power Pivot, you can:
 - Work with large data sets
 - Build extensive relationships
 - Create complex (or simple) calculations
- All of this is done within a high-performance environment and the familiar interface of Excel.

Start the Power Pivot add-in for Excel

1. Go to **File > Options > Add-Ins**.
2. In the **Manage** box, click **COM Add-ins > Go**.
3. Check the **Microsoft Office Power Pivot** box, and then click **OK**. If you have other versions of the Power Pivot add-in installed, those versions are also listed in the COM Add-ins list. Be sure to select the Power Pivot add-in for Excel.



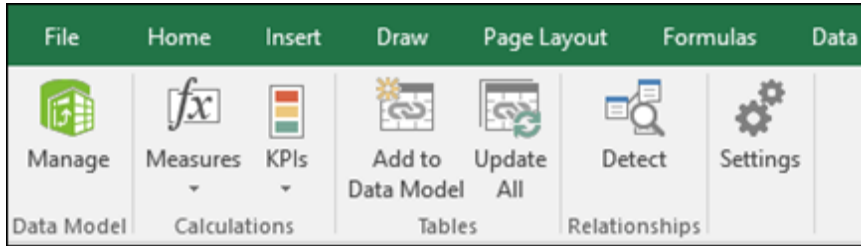
Open the Power Pivot window

1. Click **Power Pivot**.
2. This is the tab where you work with Power Pivot PivotTables, calculated fields, and key performance indicators (KPIs), and creating linked tables.
3. Click **Manage**.

Now you're in the Power Pivot window. Here you can click Get External Data to use the Table Import Wizard to filter data as you add it to your file, create relationships between tables, enrich the data with calculations and expressions, and then use this data to create PivotTables and PivotCharts.



Getting Started with Power Pivot



Power Pivot for Excel - Book1

File Home Design Advanced

Paste Append Paste Replace Copy Clipboard

From Database From Data Service From Other Sources Get External Data

[Order ID] fx

	Order Date	Sales Rep	Customer N...
1	1/7/2014	Anne Hellung-Lar...	Company F
2	1/12/2014	Robert Zare	Company J
3	1/15/2014	Jan Kotas	Company AA
4	1/20/2014	Mariya Sergienko	Company D

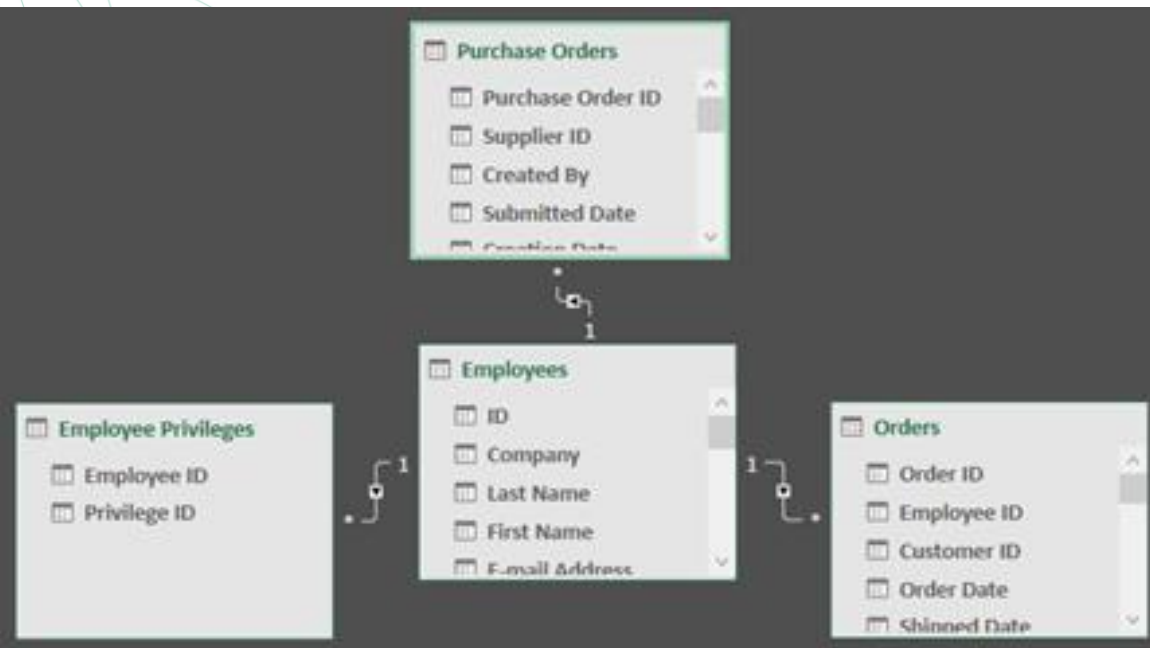
When the **Power Pivot add-on** is enabled, the **Power Pivot** tab in the ribbon is available, as shown in the following image.

From the **Power Pivot** ribbon tab, select **Manage** from the **Data Model** section.

When you select **Manage**, the **Power Pivot window** appears, which is where you can view and manage the data model, add calculations, establish relationships, and see elements of your Power Pivot *data model*. A **data model** is a collection of tables or other data, often with established relationships among them. The following image shows the **Power Pivot window** with a table displayed.

Getting Started with Power Pivot

The **Power Pivot window** can also establish, and graphically represent, relationships between the data included in the model. By selecting the **Diagram view** icon from the bottom right side of the **Power Pivot window**, you can see the existing relationships in the Power Pivot data model. The following image shows the **Power Pivot window** in **Diagram view**.



Publishing in the Power BI service

Publishing to Power BI

✓ Success!

[Open 'Contoso Product Line.pbix' in Power BI](#)

[Get Quick Insights](#)

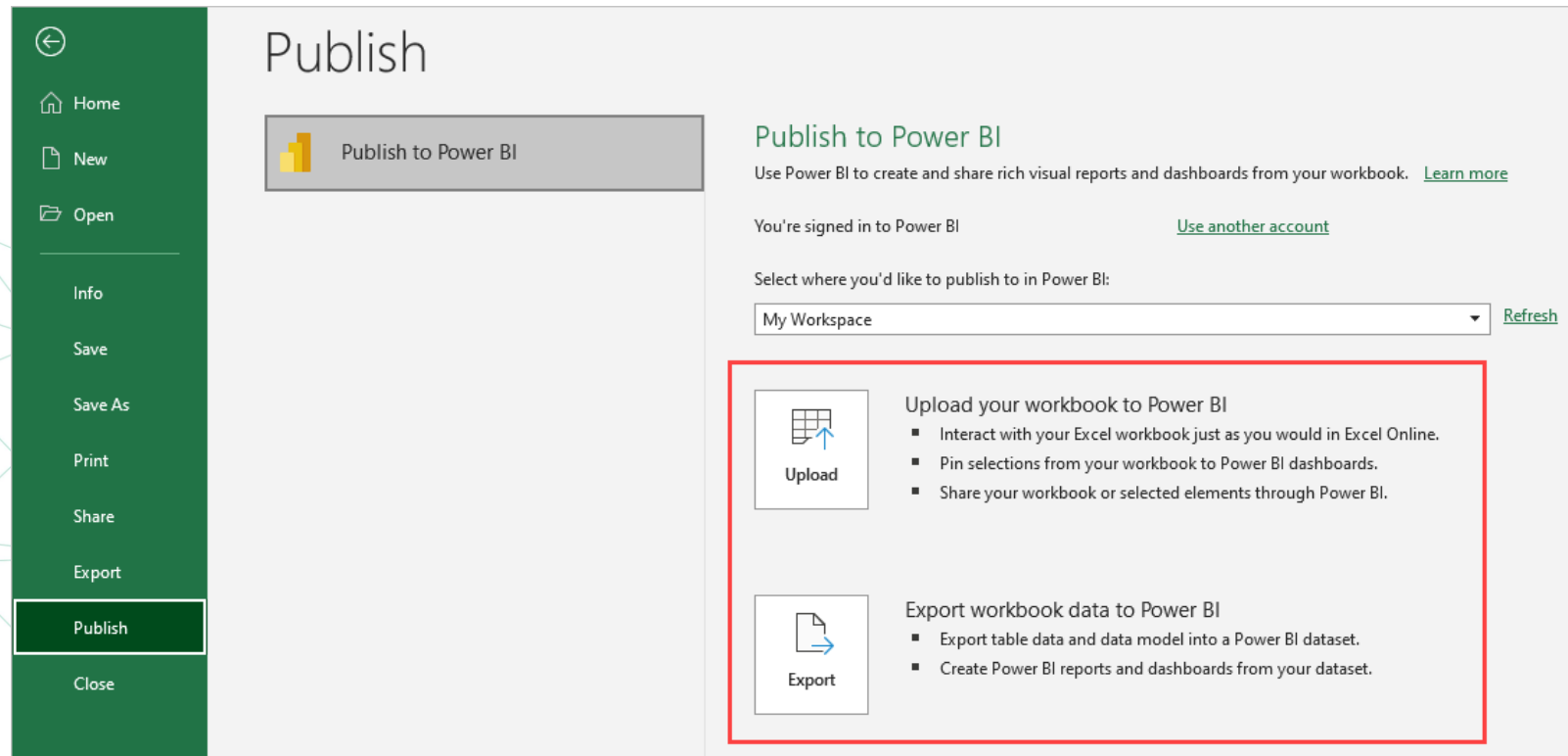
Publishing in **Power BI** is the process of sharing reports and dashboards with others. This is typically done through the Power BI service, where reports are hosted.

To publish a report in Power BI:

- Third select the link to open the report in the Power BI site or right-click to share it.

Publishing in the Power BI service

To publish your Excel workbook to Power BI, in Excel, select **File** > **Publish** and select either **Upload** or **Export**. The following screenshot shows the two options for how to get your workbook into Power BI:



- **Considerations when publishing in the Power BI service**

Data source availability

Security and permissions

Version control and backup

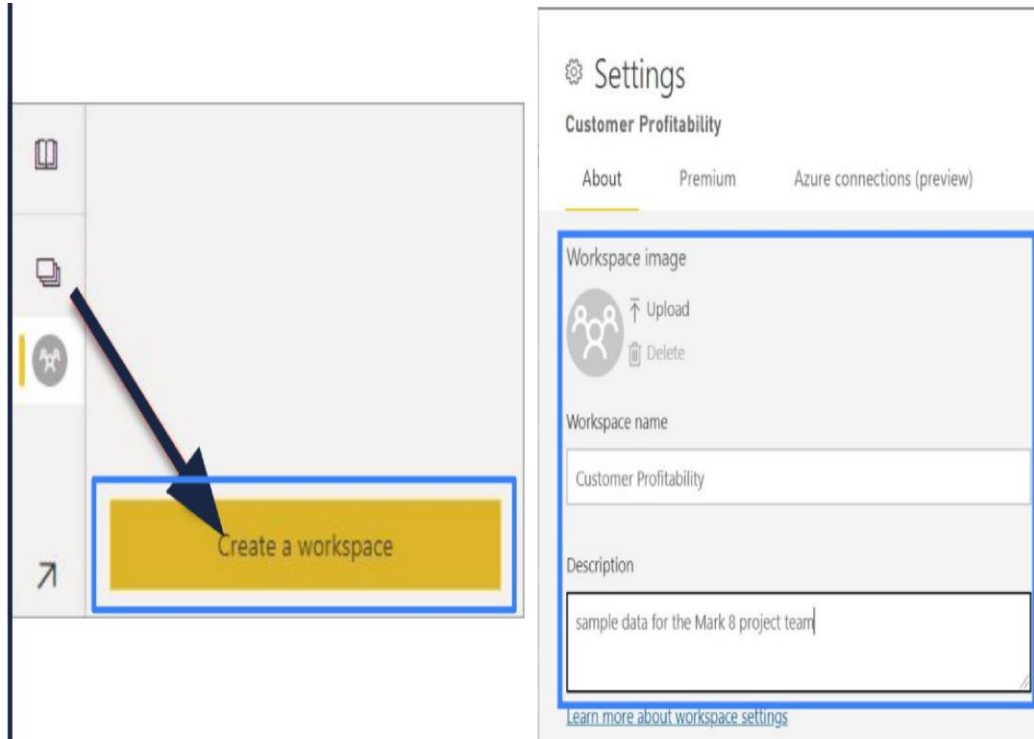
- **Considerations when publishing in the Power BI service**

Data refresh and schedule

Report design and layout

Testing and validation

Creating Workspaces in the Power BI Service



Creating Workspaces in the Power BI Service

Workspaces are containers used for organizing and managing reports. They are essential for collaboration, content sharing, and access control within the Power BI service. Workspaces allow you to group related reports and dashboards for specific teams, projects, or purposes.

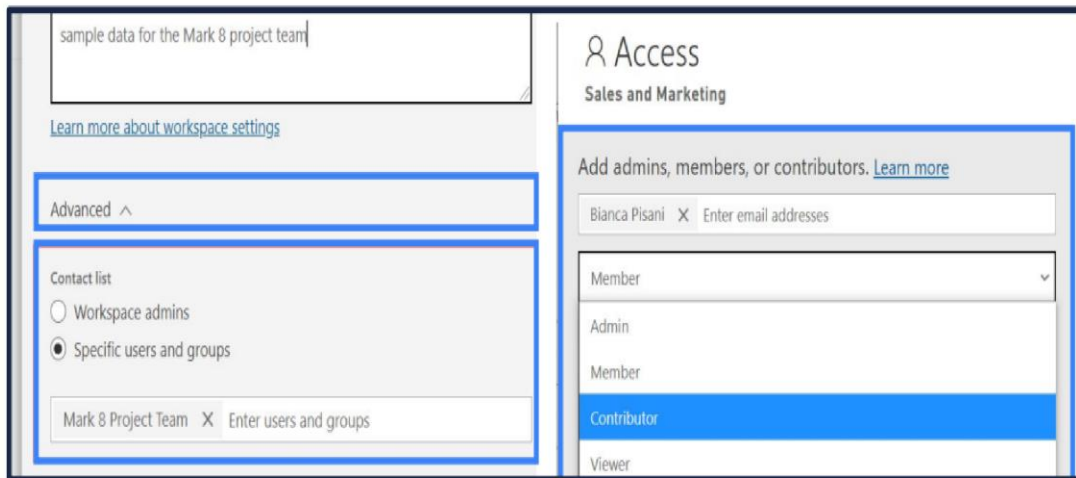
To create a workspace in the Power BI service:

1. In the Power BI service, go to the **"Workspaces"** tab.
2. Select **"Create a workspace"**.
3. Provide a **name** and **description** for the workspace.

Creating Workspaces in the Power BI Service

Creating Workspaces in the Power BI Service

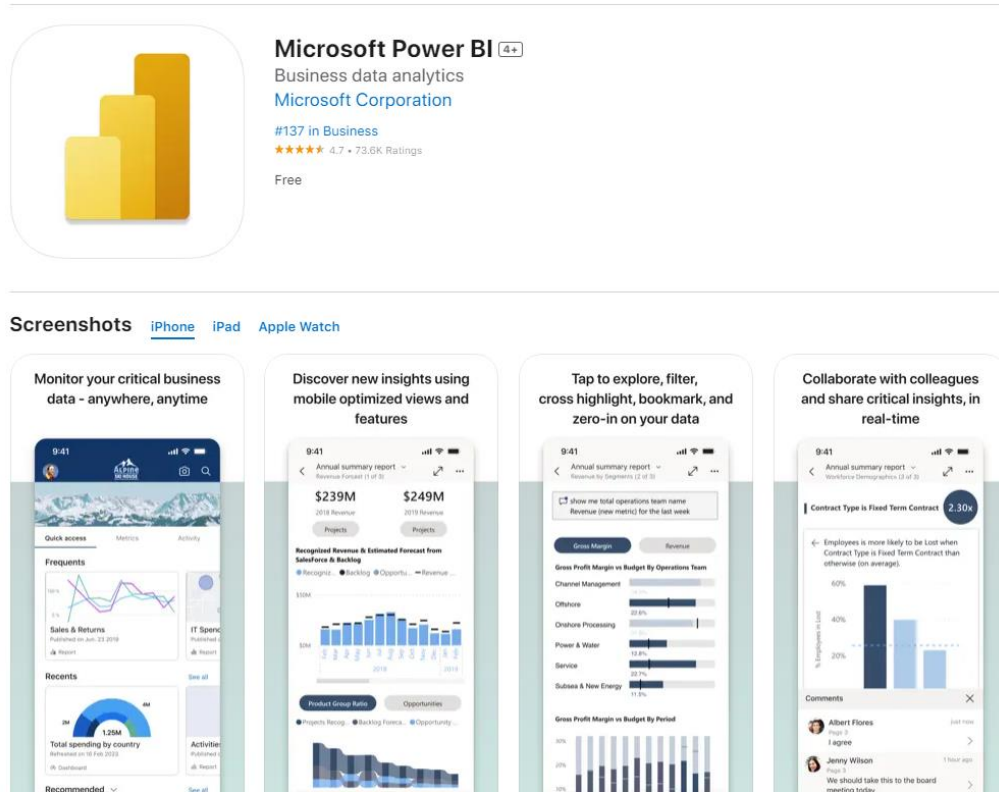
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To create a workspace in the Power BI service:

1. In the Power BI service, go to the **"Workspaces"** tab.
2. Select **"Create a workspace"** and provide a **name** and **description** for the workspace.
3. Invite members to join the workspace and **assign roles**. These roles determine the actions members can perform within the workspace.

Sharing via the Power BI Mobile App

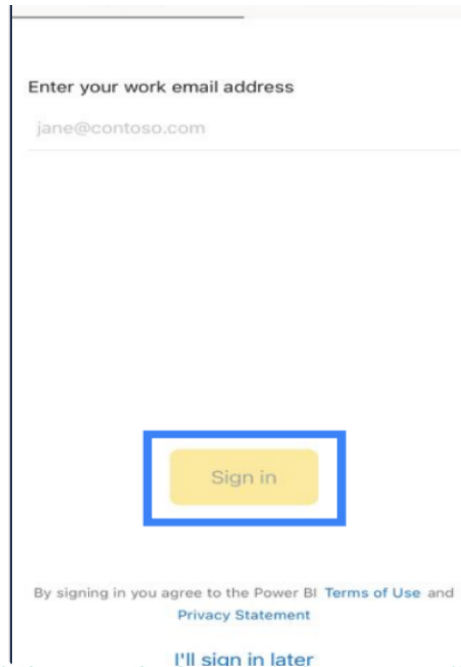


Users can access and interact with reports and dashboards on their mobile devices using the Power BI mobile app. This app functions similarly to the Power BI service, but on a mobile platform.

To share via the mobile app:

- First download and install the Power BI mobile app on an iOS or Android device.

Sharing via the Power BI mobile app



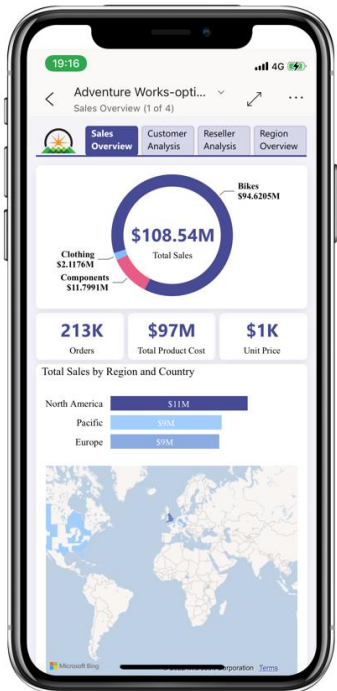
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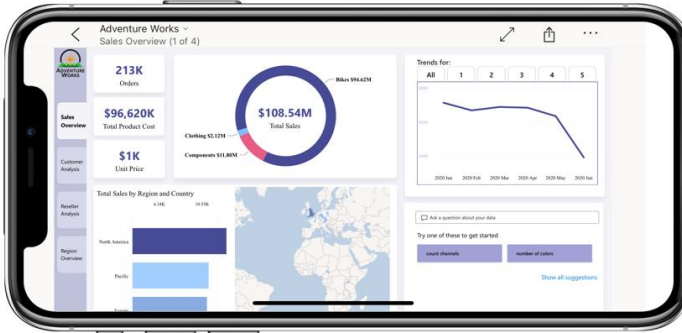
- Second sign in to the app using our Power BI account credentials.

Sharing via the Power BI mobile app

Mobile-optimized view



Regular, non-optimized view

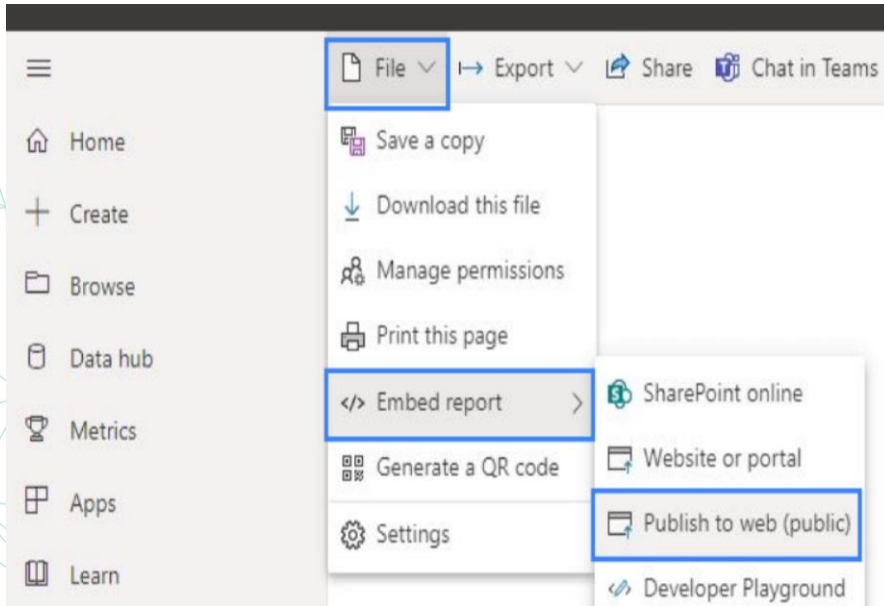


Users can access and interact with reports and dashboards on their mobile devices using the Power BI mobile app. This app functions similarly to the Power BI service, but on a mobile platform.

To share via the mobile app:

- Third access reports and dashboards shared via workspaces in the app.

Publish to Web



The "**Publish to web**" feature enables you to share your reports publicly on the internet without requiring viewers to sign in.

To publish to web in the Power BI service:

1. In the Power BI service, navigate to the report you want to publish.
2. Select "**File**", then "**Embed report**", and choose "**Publish to web**".
3. The **Publish to web** feature will provide an **embed code** or **link** that can be shared with others to allow public access to the report.

This feature allows you to share your reports with a wider audience, including those without Power BI accounts, making your data and insights accessible to the public.

• Considerations when publishing to web

"Publish to web" is a useful method for sharing Power BI reports, but it's important to exercise caution particularly when it comes to data security and sensitivity.

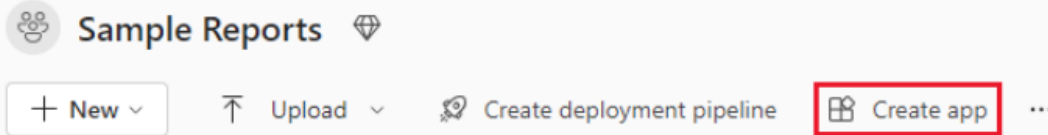
Data sensitivity

Expiration and Data refresh

Limitations

Building and Using Apps in the Power BI Service

Power BI apps are bundled collections of reports, dashboards, and datasets tailored for specific audiences. They enhance the user experience, streamline content sharing, and provide structured insights for organizations.



To build an app in Power BI:

1. In the workspace list view, select **“Create app”** to start the process of creating and publishing an app from the workspace.
2. Follow the steps to configure and publish your app.

Note that the **“Publish to web”** feature is not part of app creation but is used to share reports publicly. Ensure you use the app-building features for audience-specific access and content structuring.

Building and Using Apps in the Power BI Service

① Setup* ② Content* ③ Audience*

Build your app

App name *

Sample Reports App

Description *

Enter a summary

Describe your app.

App logo

 Upload
Delete

App theme color



Contact information

☒ Show app publisher
☐ Show items contacts from the workspace
☐ Show specific individuals or groups

Enter a name or email address

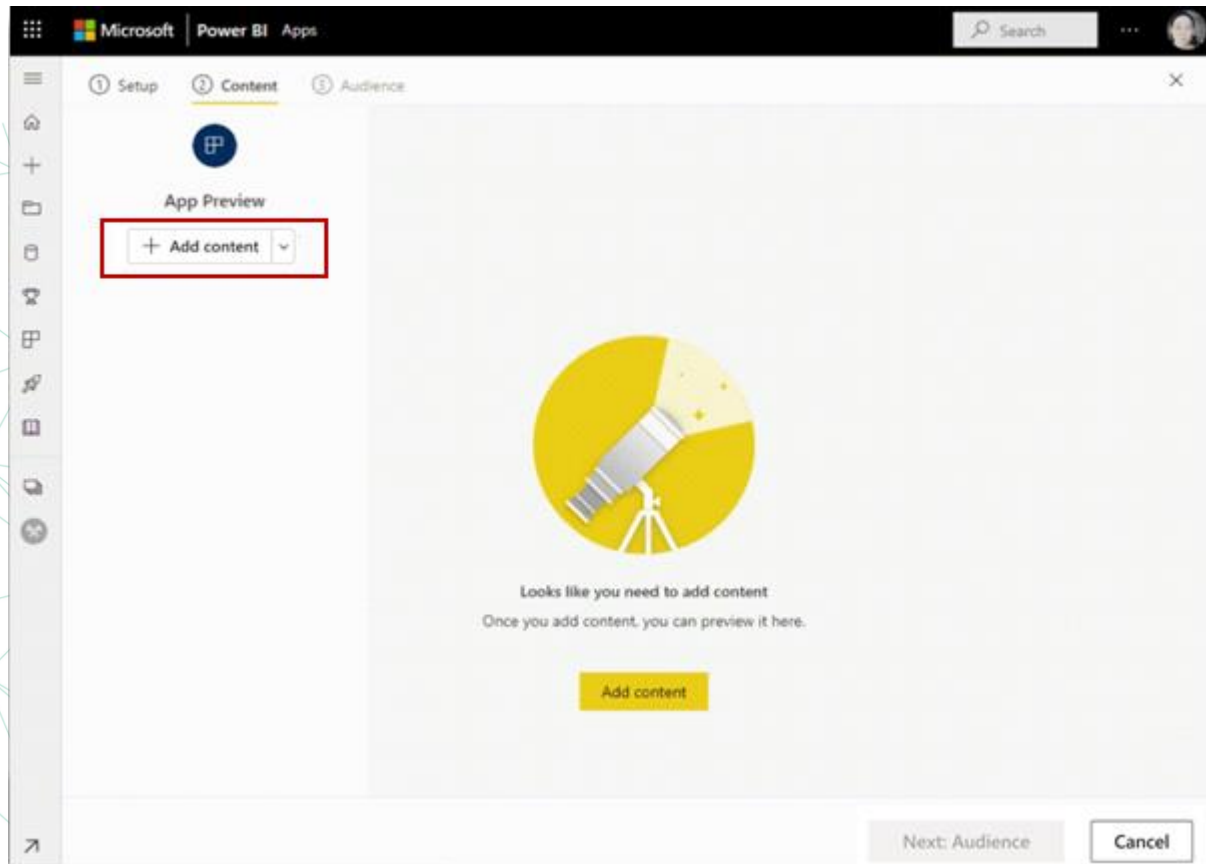
Advanced settings ▾

To build an app in Power BI:

Create the App: Start the process by selecting “**Create app**” in the workspace list view.

Customize the App: Customize the appearance of the app, including its **name**, **logo**, **colors**, and **branding**. This helps create a consistent and user-friendly experience.

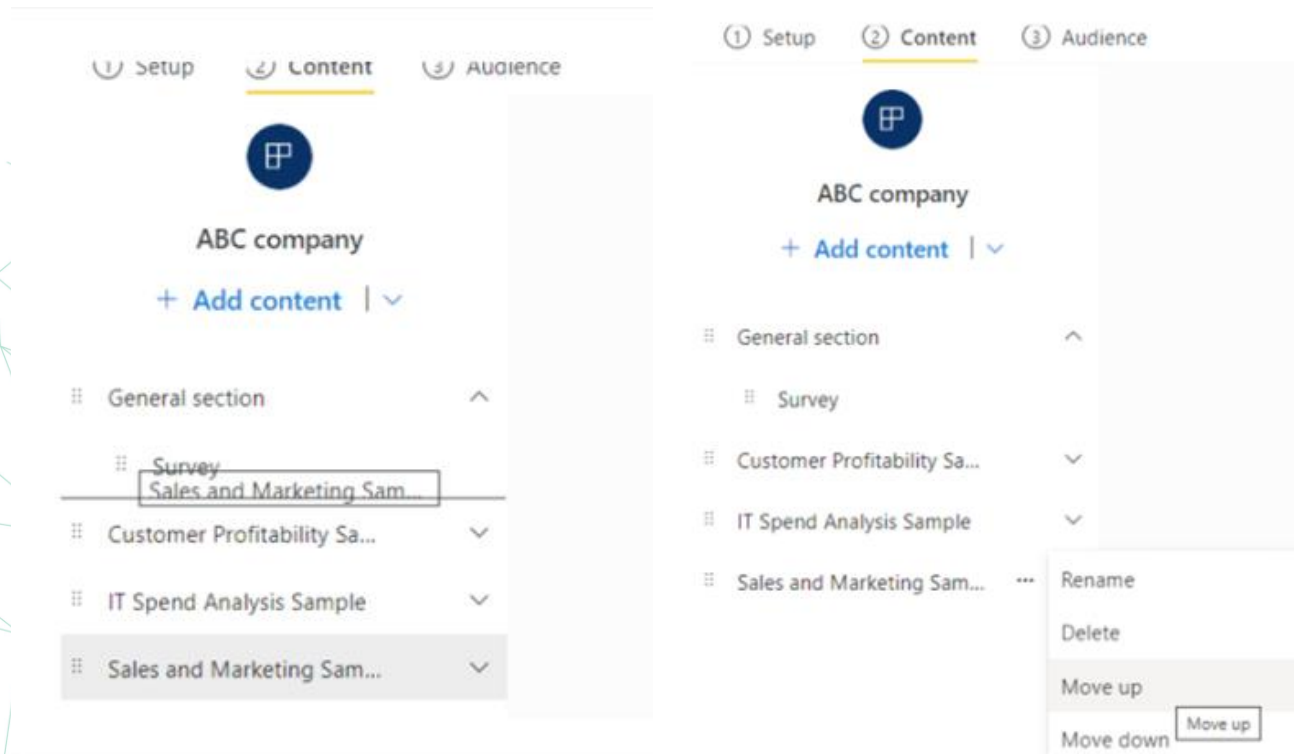
Building and Using Apps in the Power BI Service



To build an app in Power BI:

Third select the relevant content that we would like to add from the current workspace.

Building and Using Apps in the Power BI Service

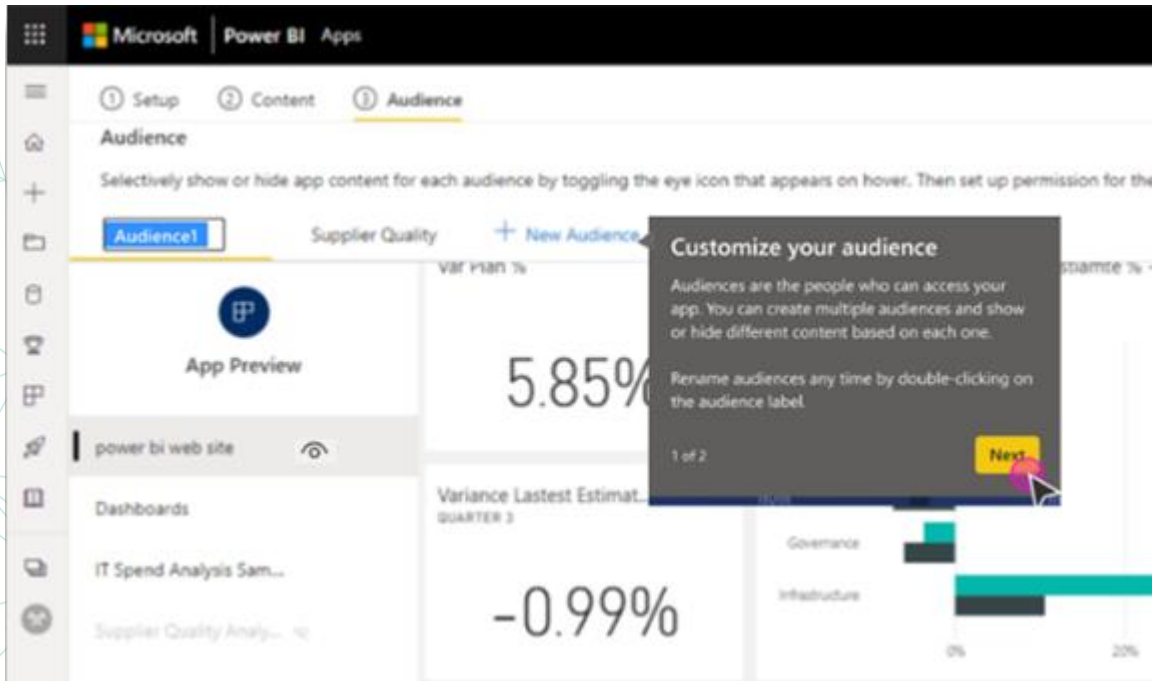


Drag the content up or down in the list.

Select the Move up or Move down options next to each item.

- To build an app in Power BI: **Furth** organize the content within the app workspace to create a logical navigation structure using the Power BI app navigation builder to design menus, sections, and landing pages.

Building and Using Apps in the Power BI Service



Fifth the next step in creating a Power BI app is to define the audience, which are multiple groups with differing access to content.

Sixth we then manage each audience's access and add or conceal content from them to make it unavailable or visible in the published app.

Finally, we can publish the app and copy and share the link.

How Users Can Interact with the App

Users can access published apps in the "**Apps**" section of the Power BI service or find them in the Power BI mobile app.

Within the app, users can:

- **Explore Content:** Navigate through various reports, dashboards, and datasets guided by the app's structure and navigation.
- **Interact with Visuals:** Engage with visualizations to gain insights.
- **Apply Filters:** Refine the data displayed by applying filters.
- **Drill into Data:** Dive deeper into data for more detailed analysis.

This interaction allows users to fully leverage the insights and functionality provided by the app.

How users can interact with the app

- **The app may include collaboration features such as:**
- **Comments:** Users can engage in discussions directly within the app.
- **Email Subscriptions:** Users can subscribe to receive updates and notifications.
- **Additionally:**
- **Mobile Access:** Users can access app content on mobile devices using the Power BI mobile app, providing convenience for on-the-go insights.

Exporting and Printing Dashboards and Reports

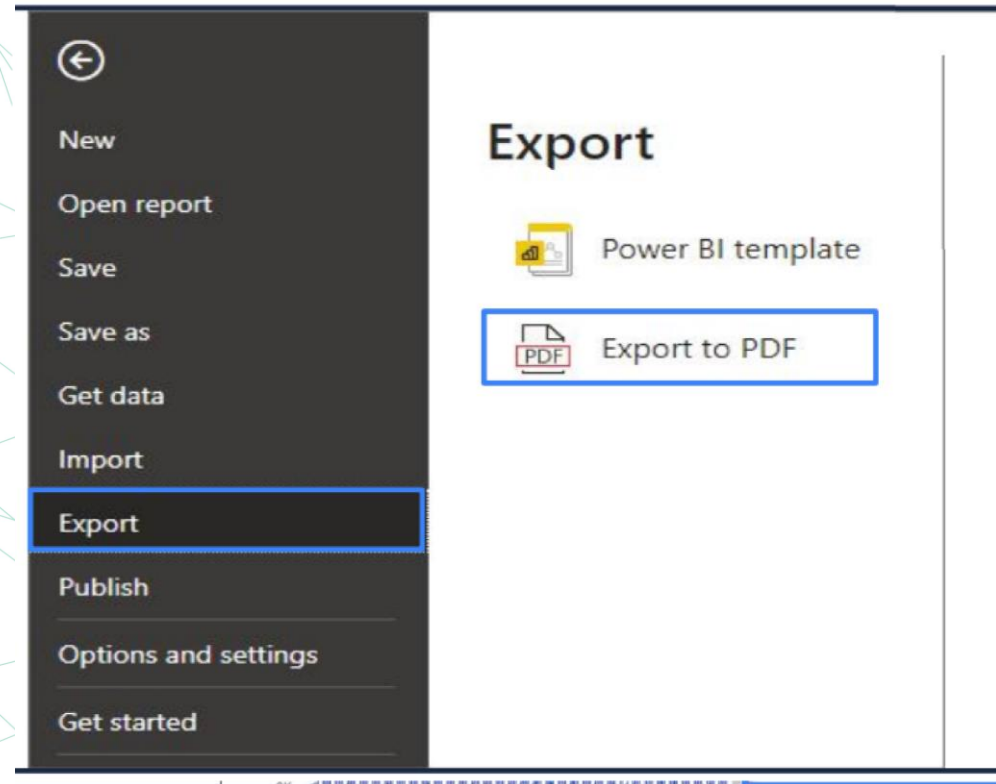
Power BI allows you to export and print your reports and dashboards in various formats, making it easier to share information with others.

To export a report:

1.Open the Report: Open the report or dashboard you wish to export.

2.Navigate to the Export Option: Go to the "File" menu and select the "Export" option.

3.Choose the Format: Select the desired format for export, such as **PDF**, **PowerPoint**, **Excel**, or others.



Q&A

Task

- **Power Pivot**

- **Task:**

- Use **Power Pivot** to build a data model using **multiple tables** (e.g., Sales, Customer, and Product data).
- Establish **relationships** between the tables based on common fields (e.g., Customer ID, Product ID).
- Add **calculated fields** to compute total revenue and profit.
- Analyze the **profitability of different customer segments** (e.g., by region, category).

- **Challenge:**

- Perform an **advanced analysis** using Power Pivot to identify the **top 5 most profitable products** and determine which regions contribute the most to overall profitability.

Thank You